**CSE 332: Computer Architecture and Organization**
**Assignment#2**
**Spring 2025**

**Name: ___________________________________**
**Student ID: ___________________________**

**Direction:** **You can consult any resources such as books, online references, and videos for this assignment, however, you have to properly cite and paraphrase your answers when it is necessary. There will be points for partial attempts. You can upload a typed or handwritten copy of your assignment to the canvas. However, you have to show the original copy of your assignment in case of arising questions about the authenticity. You should upload your assignment to canvas by the due date.**

**Submission guidelines:** **All our assignments will be based on 100 points so that we can assign weights to get your points for final grade. Whoever fails to submit within the time period assigned through NSU canvas but submit the assignment within the next three days will be punished by 20% deduction of points (e.g. start with -20 points within 100 points). The late submission will be open until semester's end through canvas. After the first 2 days of grace period, the deduction will be 40%.**

**Note:** **You are allowed to use only instructions implemented by the actual MIPS hardware provided in the Canvas references. Use assembly language format from the references or the book. Note, base ten numbers are listed as normal (e.g. 23), binary numbers are prefixed with 0b or format such as $XX_2$ and hexadecimal numbers are prefixed with 0x or format such as $XX_{16}$/$XX_{HEX}$**

1. **(30 points, 5 each)** What is the single MIPS instruction, or if not possible, the shortest sequence of MIPS instructions that performs:
    1. Move content of register r11 into register r23.
    2. Load register r15 with the value 265,140.
    3. Store the value 55,259 into a word at memory location 40000.
    4. The equation a = b + c,assuming a, b and c are integer variables at memory location 0b10000000, 0b1001000 and 0b100010011100, respectively.
    5. The equation y[5] = y[10] + x, assuming x corresponds to register r2, and y[1] is an integer at the memory location 524,292 is the first element of vector y.

6. The statement: if (a <= b) goto lab1, assuming that variable a corresponds to the word at memory address 400, and variable b corresponds to register r10. The instruction with lab1 is before the branch instruction and there are 20 (machine) instructions between those two instructions.

2. **(20 points)** For the following MIPS program determine the bit pattern for each instruction.

```
lap:    lw      r16, 0(13)
        addi    r8, r2, -513
        nop
        sw      r8, -18(r4)
        bne     r3, r0, lap    %lap= 25
```

3. **(30 points)** Write an assembly program in MIPS32 ISA for the following C program.
4.

```
#define N=11;

int ARRAY[N]= {1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21};

void main (void) {
        int total = sum_array(ARRAY);
        printf ("The total is: %d", total);
}
int sum_array(int ARRAY[N]) {
        Int total=0;
            for (int i=0; i<N; i++ ) {
                total+ =ARRAY[i];
        }
        return total;
 }
```